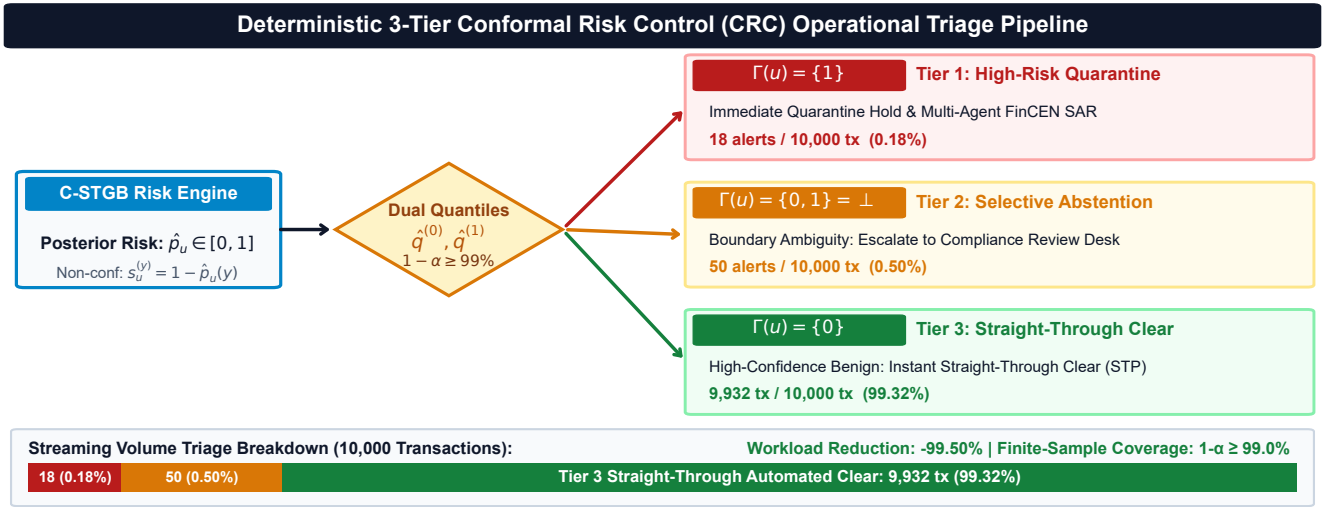

Deterministic 3-Tier Conformal Risk Control (CRC) Operational Triage Pipeline
C-STGB Risk Engine
Posterior Risk: $\hat{p}_u \in [0, 1]$
Non-conf: $s_u^{(y)} = 1 - \hat{p}_u(y)$
Dual Quantiles
$\hat{q}^{(0)}, \hat{q}^{(1)}$
$1 - \alpha \geq 99\%$
$\Gamma(u) = \{1\}$
Tier 1: High-Risk Quarantine
Immediate Quarantine Hold & Multi-Agent FinCEN SAR
18 alerts / 10,000 tx (0.18%)
$\Gamma(u) = \{0, 1\} = \perp$
Tier 2: Selective Abstention
Boundary Ambiguity: Escalate to Compliance Review Desk
50 alerts / 10,000 tx (0.50%)
$\Gamma(u) = \{0\}$
Tier 3: Straight-Through Clear
High-Confidence Benign: Instant Straight-Through Clear (STP)
9,932 tx / 10,000 tx (99.32%)
Streaming Volume Triage Breakdown (10,000 Transactions):
Workload Reduction: -99.50% | Finite-Sample Coverage: 1-α ≥ 99.0%
18 (0.18%)
50 (0.50%)
Tier 3 Straight-Through Automated Clear: 9,932 tx (99.32%)